

```
1 import java.util.Scanner;
2
3 public class Main {
4
5     public static void main(String[] args) {
6         Scanner scanner = new Scanner(System.in);
7         Zoo myZoo = new Zoo("My Awesome Zoo");
8
9         // Attempt to load previously saved data on
startup
10        myZoo.loadData();
11
12        boolean isRunning = true;
13
14        while (isRunning) {
15            System.out.println("\n--- Zoo Management
System ---");
16            System.out.println("1. Add a new animal"
);
17            System.out.println("2. Remove an animal"
);
18            System.out.println("3. Search for an
animal");
19            System.out.println("4. Modify an animal's
weight");
20            System.out.println("5. Display all
animals");
21            System.out.println("6. Perform daily care
routines");
22            System.out.println("7. Print zoo report"
);
23            System.out.println("8. Feed all animals"
);
24            System.out.println("9. Calculate total
weight");
25            System.out.println("10. Save and Exit");
26            System.out.print("Select an option: ");
27
28            String choice = scanner.nextLine().trim
();
29
```

```
30         switch (choice) {
31             case "1":
32                 addAnimalMenu(scanner, myZoo);
33                 break;
34             case "2":
35                 System.out.print("Enter the name
of the animal to remove: ");
36                 String nameToRemove = scanner.
nextLine().trim();
37                 myZoo.removeAnimal(nameToRemove);
38                 break;
39             case "3":
40                 System.out.print("Enter name or
colour to search: ");
41                 String query = scanner.nextLine
().trim();
42                 myZoo.searchAnimal(query);
43                 break;
44             case "4":
45                 System.out.print("Enter the name
of the animal to modify: ");
46                 String modName = scanner.nextLine
().trim();
47                 System.out.print("Enter new
weight: ");
48                 try {
49                     double newWeight = Double.
parseDouble(scanner.nextLine().trim());
50                     myZoo.modifyAnimalWeight(
modName, newWeight);
51                 } catch (NumberFormatException e
) {
52                     System.out.println("Error:
Please enter a valid number for weight.");
53                 }
54                 break;
55             case "5":
56                 myZoo.displayAllAnimals();
57                 break;
58             case "6":
59                 myZoo.performDailyCare();
```

```

60             break;
61         case "7":
62             myZoo.printZooReport();
63             break;
64         case "8":
65             myZoo.feedAnimals();
66             break;
67         case "9":
68             myZoo.calculateTotalWeight();
69             break;
70         case "10":
71             myZoo.saveData();
72             isRunning = false;
73             System.out.println("Data saved.
    Exiting the system. Goodbye!");
74             break;
75         default:
76             System.out.println("Invalid
    option. Please try again.");
77     }
78 }
79 scanner.close();
80 }
81
82 private static void addAnimalMenu(Scanner
scanner, Zoo zoo) {
83     System.out.println("\n--- Select Animal Type
    ---");
84     System.out.println("1. Eagle");
85     System.out.println("2. Penguin");
86     System.out.println("3. Duck");
87     System.out.print("Type: ");
88
89     String typeChoice = scanner.nextLine().trim
    ();
90
91     // Blank Name Check
92     System.out.print("Enter name: ");
93     String name = scanner.nextLine().trim();
94     if (name.isEmpty()) {
95         System.out.println("Error: Animal name

```

```

95 cannot be blank. Aborting.");
96         return;
97     }
98
99     // Age Validation Loop
100    int animalAge = -1;
101    while (animalAge < 0) {
102        System.out.print("Enter animal age: ");
103        try {
104            animalAge = Integer.parseInt(scanner
105                .nextLine().trim());
106            if (animalAge < 0) {
107                System.out.println("Error: Age
108                    cannot be negative. Please try again.");
109            }
110        } catch (NumberFormatException e) {
111            System.out.println("Error: Please
112                enter a real number.");
113            animalAge = -1;
114        }
115    }
116
117    System.out.print("Enter colour: ");
118    String colour = scanner.nextLine().trim();
119
120    // Weight Validation Loop (Extra protection
121    !)
122    double animalWeight = -1.0;
123    while (animalWeight < 0) {
124        System.out.print("Enter animal weight (
125            kg): ");
126        try {
127            animalWeight = Double.parseDouble(
128                scanner.nextLine().trim());
129            if (animalWeight < 0) {
130                System.out.println("Error:
131                    Weight cannot be negative. Please try again.");
132            }
133        } catch (NumberFormatException e) {
134            System.out.println("Error: Please
135                enter a real number.");

```

```

128         animalWeight = -1.0;
129     }
130 }
131
132     // Create the specific animal
133     Animal newAnimal = null;
134     try {
135         if (typeChoice.equals("1")) {
136             System.out.print("Enter wingspan (m
137 ): ");
138             double wingspan = Double.parseDouble
139 (scanner.nextLine().trim());
140             newAnimal = new Eagle(name,
141 animalAge, colour, animalWeight, wingspan);
142         } else if (typeChoice.equals("2")) {
143             System.out.print("Enter swim speed (
144 km/h): ");
145             double swimSpeed = Double.
146 parseDouble(scanner.nextLine().trim());
147             newAnimal = new Penguin(name,
148 animalAge, colour, animalWeight, swimSpeed);
149         } else if (typeChoice.equals("3")) {
150             System.out.print("Enter beak colour
151 : ");
152             String beak = scanner.nextLine().
153 trim();
154             newAnimal = new Duck(name, animalAge
155 , colour, animalWeight, beak);
156         } else {
157             System.out.println("Error: Invalid
158 animal type selected.");
159             return;
160         }
161     }
162
163     // Add to zoo
164     zoo.addAnimal(newAnimal);
165
166 } catch (NumberFormatException e) {
167     System.out.println("Error: Invalid
168 numeric input for specific trait. Aborting.");
169 }

```

```
158     }
```

```
159 }
```